Mohamed Amin Makni

Software Engineer | AI, Computer Vision & Full-Stack Development

Sfax, Tunisia | +216 52 855 085 | mki.medamin@gmail.com | linkedin.com/in/makni-med-amin | github.com/medaminmakni | med-amin-makni.vercel.app

Experience

AI Engineer Intern (Final-Year Project) | *Essilor SIVO (EssilorLuxottica)* | *Sfax, Tunisia* **Feb 2026 – Jul 2026**

- Developed the document-processing pipeline and a **React Native** app for opticians to photograph order forms, review extracted data, submit orders and track their status.
- Trained a **YOLO11** model to locate form fields, reaching **95.16% mAP@50**, and an **EfficientNetV2-S** classifier, both in **PyTorch**.
- Extracted text with **TrOCR** and **GLM-OCR**, then validated each field with business rules and fuzzy matching (**RapidFuzz**).
- Used confidence thresholds to send uncertain predictions for human review before saving the extracted data.
- Gathered requirements with business teams, prepared training data, evaluated models and integrated the system with the backend.

Backend Developer Intern | *Altavo Partners* | *Remote (Paris, France)* **Jul 2025 – Aug 2025**

- Built the bank and arbiter side of Hex-Port, a Hedera-based trade-finance platform for African exporters: KYC review, document validation, disputes and bank authentication.
- Developed REST APIs in **TypeScript** with **Node.js**, **Express** and **PostgreSQL** (**Prisma**), including the approval flow that releases escrow at shipment and delivery.
- Recorded every order state change on a Hedera consensus topic (**Hedera SDK**), giving a verifiable audit trail; worked alongside **Solidity** smart contracts.

Full-Stack Developer Intern | *Infotech Consulting Services (ICS)* | *Sfax, Tunisia* **Jul 2024 – Aug 2024**

- Developed the REST endpoints of the exams and teaching-assignment module of a university administration platform (sessions, invigilation, teaching load, timetables, registrations) with **Laravel** and **MySQL**.
- Wrote the API documentation used by the development team, on a **Laravel** / **Vue.js** stack.

Projects

SmartWarehouse AI | *Python, FastAPI, Next.js, YOLO, ChromaDB, SQL, Docker* **2025**

- Built an intelligent warehouse-management system combining computer vision, **LLM reasoning** and **RAG** to identify arriving vehicles, match them to orders and recommend loading docks.
- Trained a **YOLO** detector for Tunisian licence plates and combined **Gemini OCR**, SQL order data and RAG-based warehouse rules in a **FastAPI** and **Next.js** application.

ORYS (team of 4, AI layer lead) | *TypeScript, OpenAI & Anthropic SDKs, Zod, Vitest* **2026 – Present**

- Develop the AI assistant infrastructure for a business management SaaS within a team of four, allowing the same agent code to use different model providers.
- Built **OpenAI** and **Anthropic** adapters, tool calling with **Zod** argument validation, and token and cost tracking; tested provider behaviour and execution with **Vitest**.

Fytrak (solo) | *React Native (Expo), Firebase, Cloud Functions, TypeScript* **2026**

- Built a coaching platform to bring training plans, nutrition and coach–trainee follow-up into one app, independently developing the **React Native** app, admin console and **Firebase** backend.
- Connected workout and meal-plan assignments to trainees' activity logs and progress, giving coaches daily reports and in-app messaging to review sessions and follow up.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, PHP
AI & ML: PyTorch, TensorFlow, scikit-learn, YOLO, OCR (TrOCR, GLM-OCR), RAG, LLM agents (OpenAI, Anthropic), ChromaDB
Backend: FastAPI, Node.js, Express, Prisma, Laravel, REST APIs
Frontend & Mobile: React, Next.js, React Native (Expo), Vue.js
Data: PostgreSQL, MySQL, MongoDB, Firestore, pandas
Cloud & DevOps: Docker, Git, CI/CD, Linux, Google Cloud (Firebase, Cloud Functions)

Education

National Engineering Degree in Computer Science | *IIT Sfax* | *Tunisia* **2023 – 2026**

- Specialization: Software Engineering & Business Intelligence. Graduated July 2026.

Preparatory Classes for Engineering Schools | *IPEIS Sfax* | *Tunisia* **2021 – 2023**

Certifications & Additional Information

Certifications: Information Technology Specialist (ITS) – Artificial Intelligence (Certiport, 2024, 889/1000); ITS – Python (Certiport, 2024, 880/1000); Hashgraph Developer (The Hashgraph Association, 2025)
Languages: Arabic (native), French (fluent), English (fluent), German (basic)
Other: SIVP-eligible (Tunisia); open to relocation